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1 Paper's Focus

- What moral duties do ML companies have when producing their models?
- Who would be acting unethically for not taking responsibility when the models cause damange?
- Is it ethically feasible to enforce regulations in the black box that are ML model?

2 Background Information

The issue that "Coded Bias" is trying to tackle is that of algorithms and how training them can cause them to have biases against minority groups misrepresented in the used data training sets.

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2.1 Stakeholders

There are several stakeholders in this scenario with a major power imbalance between the machine learning companies and the general public:

- Machine Learning Companies: As they decide what gets made and how it is made
- Machine Learning Engineers: As they implement the algorithms and methods of training
- Governments: As they regulate and use the algorithms for various purposes
- Software companies: As they use the algorithms to enhance their businesses
- General public: As they are the ones who are being judged by the algorithm

This power imbalance places the general public at a major disadvantage as machine learning algorithms rely on pre-existing data which is not guaranteed to be sanitized of biases. Therefore causing unjust profiling of individuals, rejections in jobs because of nationality/ethnicity, and more without any form of control as the mathematical models are not easily reverse engineerable and companies have no incentive to do so.

3 Documentaty's purpose

The documentary is aiming to bring to attention the lack of accountability machine learning algorithms and their users are facing in terms of the impact they have in the general public due to the biases. The documentary is posing a solution in terms of regulation in the usage and implementaiton of the algorithms to reduce discrimination.

4 Technological Background

The focus of this documentary is on machine learning algorithms therefore to be able to properly analyze them there is a need to understand the following:

- Data Collection & Cleansing
  - Data collection article [Vidhya et al. Jun]
  - Data cleaning article [Lee et al. Sep]
- Types of Machine Learning [China NA]:
  - Supervised learning
  - Unsupervised learning
  - Self-supervised learning
  - Reinforced learning
  - Semi-supervised learning
- Machine Learning Algorithms (brief math overview) [Bergmann NA]

By understanding all of this, it is possible to evaluate the type of control that companies that produce and sell these algorithms have over the biases that are being baked into their software.

5 Ethical Analysis

5.1 Responsibilities of ML companies

5.2 Kantianism

5.3 Act Utilitarianism

5.4 Responsibilities of ML companies

5.5 Social Contract Theory

5.6 Act Utilitarianism

5.7 Ethical feasibility of creating regulations

6 Beyond the superficial

A lot of DigiPen graduates tend to go out into the world and work in embedded or real-time systems that rely on ML to produce the adequate output, be it a sensor for a car, or a self-control algorithm for a plane. It is very important to understand how the usage of ML can bias the algorithms in ways we can't control as the models are trained beyond our reach.

References

Dave Bergmann. N/A. .

Chrystal R. China. N/A. .

Ga Young Lee, Lubna Alzamil, Bakhtiyar Doskenov, and Arash Termehchy. Sep . (Sep). doi:arXiv:2109.07127

Gohri Vidhya, D Nirmala, and T Manju. Jun . (Jun). doi:10.48185/jitc.v4i1.725